

Separable Differential Equations with Initial Conditions

Answer Key

AP Calculus AB/BC

Quick Answers

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| 1. the x-integral = x^5 | 6. the x-integral = x^3 , $y(2) = 5.477$ |
| 2. the x-integral = $2x^2$ | 7. $P(10) = 55.64$ |
| 3. the x-integral = x^2 , $y(1) = 13.591$ | 8. the y-integral = e^y , $y(\ln 5) = 1.946$ |
| 4. the y-integral = $\ln(y)$, x where $y = 48 = 4 \ln(4)$ | 9. the emptying time = 96.0 |
| 5. the x-integral = $\sin(x)$, — | 10. the x-integral = $-\cos(x)$, check of $y(0) = \frac{1}{2}$, — |
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What is verified

15 of 17 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 5, 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-7 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 17 tagged checks.

1. Separate the variables in $\frac{dy}{dx} = 5x^4y$ and integrate both sides. Write your result in the form $\ln|y| = F(x) + C$.

Solution: The right side already factors as (function of x)(function of y), so divide by y and multiply by dx :

$$\frac{dy}{y} = 5x^4 dx.$$

Integrating each side, $\int \frac{dy}{y} = \ln|y|$ and $\int 5x^4 dx = x^5$. Put the single constant on the x side.

$\ln|y| = x^5 + C$

2. Separate the variables in $\frac{dy}{dx} = \frac{4x}{y}$ and integrate both sides. Write your result in the form $\frac{1}{2}y^2 = G(x) + C$.

Solution: Multiply both sides by $y \, dx$:

$$y \, dy = 4x \, dx.$$

Now $\int y \, dy = \frac{1}{2}y^2$ and $\int 4x \, dx = 2x^2$.

$$\boxed{\frac{1}{2}y^2 = 2x^2 + C}$$

3. Solve the initial value problem $\frac{dy}{dx} = 2xy$, $y(0) = 5$. Write the particular solution $y = f(x)$, then evaluate $y(1)$ to three decimal places.

Solution: Separate and integrate:

$$\frac{dy}{y} = 2x \, dx \implies \ln |y| = x^2 + C.$$

Exponentiate: $y = Ae^{x^2}$ with $A = e^C$. The initial condition $y(0) = 5$ gives $5 = Ae^0 = A$, so $A = 5$.

$$\boxed{y = 5e^{x^2}}$$

Then $y(1) = 5e^1 = 13.5914\dots$, which rounds to

$$\boxed{y(1) \approx 13.591}$$

4. A quantity y satisfies $\frac{dy}{dx} = \frac{y}{4}$ with $y(0) = 12$ and $y > 0$. Solve the initial value problem, then find the value of x at which $y = 48$.

Solution: Separate: $\frac{dy}{y} = \frac{dx}{4}$. Integrating the left side gives $\ln |y|$ and the right side gives $\frac{x}{4} + C$, so with $y > 0$,

$$y = Ae^{x/4}, \quad y(0) = 12 \implies A = 12, \quad y = 12e^{x/4}.$$

Set $12e^{x/4} = 48$, so $e^{x/4} = 4$ and $\frac{x}{4} = \ln 4$.

$$\boxed{x = 4 \ln 4 \approx 5.545}$$

5. Separate the variables in $\frac{dy}{dx} = \frac{\cos x}{2y}$ and integrate. (a) Carry out the separation and the integration. (b) Explain why this does not determine a single function until an initial condition is given.

Solution (a): Multiply by $2y \, dx$: $2y \, dy = \cos x \, dx$. Integrating, $\int 2y \, dy = y^2$ and $\int \cos x \, dx = \sin x$.

$$\boxed{y^2 = \sin x + C}$$

Solution (b) — instructor-judged. A correct response says that solving $y^2 = \sin x + C$ for y gives *two* branches, $y = +\sqrt{\sin x + C}$ and $y = -\sqrt{\sin x + C}$, because squaring destroyed the sign. Nothing in the differential equation selects one of them; an initial condition such as $y(0) = 2$ picks the branch (and fixes C) at the same time. Full credit requires both the plus-or-minus ambiguity and the role of the initial value.

6. Solve the initial value problem $\frac{dy}{dx} = \frac{3x^2}{y}$, $y(1) = 4$, for the branch with $y > 0$. Write the particular solution, then evaluate $y(2)$ to three decimal places.

Solution: Separate: $y dy = 3x^2 dx$. Integrating, $\frac{1}{2}y^2 = x^3 + C$, so $y^2 = 2x^3 + K$ where $K = 2C$. Apply $y(1) = 4$:

$$16 = 2(1)^3 + K \implies K = 14.$$

Since $y > 0$ we take the positive root.

Then $y(2) = \sqrt{2(8) + 14} = \sqrt{30} = 5.4772\dots$, which rounds to

$$y = \sqrt{2x^3 + 14}$$

$$y(2) \approx 5.477$$

7. A colony of bacteria has mass P (in thousands of cells) after t days and satisfies $\frac{dP}{dt} = 0.08P$ with $P(0) = 25$. Find $P(10)$ to two decimal places.

Solution: Separating gives $\frac{dP}{P} = 0.08 dt$, so $\ln P = 0.08t + C$ and $P = Ae^{0.08t}$. From $P(0) = 25$ we get $A = 25$, so $P = 25e^{0.08t}$. At $t = 10$, $P = 25e^{0.8} = 55.6385\dots$

$$P(10) \approx 55.64$$

8. Solve the initial value problem $\frac{dy}{dx} = e^{x-y}$, $y(0) = \ln 3$. Write the particular solution, then evaluate $y(\ln 5)$ to three decimal places.

Solution: Write $e^{x-y} = e^x e^{-y}$; the equation is separable once you see that factorisation:

$$e^y dy = e^x dx \implies e^y = e^x + C.$$

At $x = 0$, $y = \ln 3$, so $e^{\ln 3} = e^0 + C$, i.e. $3 = 1 + C$ and $C = 2$.

Then $y(\ln 5) = \ln(5 + 2) = \ln 7 = 1.9459\dots$, which rounds to

$$y = \ln(e^x + 2)$$

$$y(\ln 5) \approx 1.946$$

9. Water drains from a tank so that the depth h (centimetres) satisfies $\frac{dh}{dt} = -\frac{3}{2\sqrt{h}}$, t in minutes, with $h = 36$ at $t = 0$. Find the time until the tank is empty.

Solution: Separate: $2\sqrt{h} dh = -3 dt$. Integrating, $\frac{4}{3}h^{3/2} = -3t + C$. At $t = 0$, $h = 36$, and $36^{3/2} = 216$, so

$$C = \frac{4}{3}(216) = 288.$$

The tank is empty when $h = 0$: $0 = -3t + 288$, so $t = \frac{288}{3}$.

$$t = 96.0$$

10. Consider $\frac{dy}{dx} = y^2 \sin x$, $y(0) = \frac{1}{2}$. (a) Find the particular solution and confirm $y(0) = \frac{1}{2}$. (b) Explain why the interval of definition stops before $x = \pi$.

Solution (a): Separate: $y^{-2} dy = \sin x dx$. Integrating, $-\frac{1}{y} = -\cos x + C$, so $\frac{1}{y} = \cos x - C$. At $x = 0$, $y = \frac{1}{2}$, so $2 = 1 - C$ and $C = -1$. Hence $\frac{1}{y} = \cos x + 1$.

$$y = \frac{1}{1 + \cos x}$$

Check the initial value: at $x = 0$, $1 + \cos 0 = 2$, so

$$y(0) = \frac{1}{2}$$

Solution (b) — instructor-judged. A correct response notes that $1 + \cos x = 0$ exactly when $x = \pi$, so the formula has a zero denominator there and $y \rightarrow \infty$ as $x \rightarrow \pi^-$. The solution through $(0, \frac{1}{2})$ therefore lives on $(-\pi, \pi)$ and cannot be continued past $x = \pi$. Full credit names the vanishing denominator and states the consequence for the interval.